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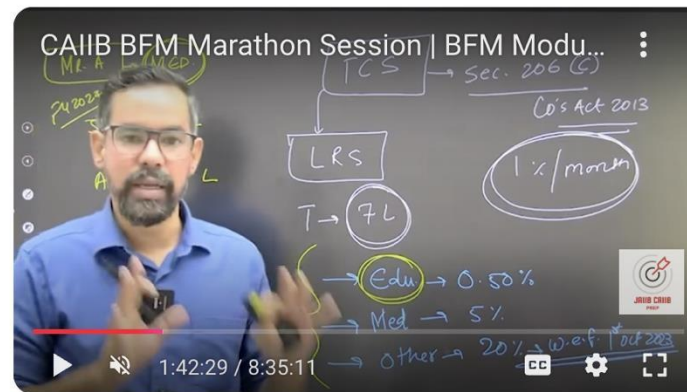
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MODULE – B

ADVANCED CONCEPTS OF FINANCIAL MANAGEMENT

- Unit 7. Sources of Finance and Financial Strategies
- Unit 8. Financial and Operating Leverages
- Unit 9. Capital Investment Decisions
- Unit 10. Capital Budgeting for International Project Investment Decisions
- Unit 11. Adjustment of Risk and Uncertainty in Capital Budgeting Decision
- Unit 12. Decision Making





SOURCES OF FINANCE AND FINANCIAL STRATEGIES

STRUCTURE

- 7.0 Objectives
- 7.1 Introduction
- 7.2 Equity Capital
- 7.3 Internal Accruals
- 7.4 Preference Capital
- 7.5 Term Loans
- 7.6 Debentures
- 7.7 Alternative Financing Strategies in the Context of Regulatory Requirements

Let Us Sum Up

Keywords



Equity Capital

Equity Capital refers to the money invested in a business by the promoter/owner, representing ownership and residual interest after liabilities are settled.

Key Features & Advantages of Equity Capital:

- Divided into Equity Shares.
- Each share has equal nominal value.
- Equity holders have two rights:
 - ✓ Right to income (dividend)
 - ✓ Right to surplus in liquidation
 - ✓ Voting rights (except in Differential Voting Right shares)

Equity Capital

Shares can be subscribed by anyone as per Articles of Association.

- Governed by Memorandum and Articles of Association.
- Transferees of shares must follow the same documents.
- Exit possible by following prescribed rules.
- Listed shares have better liquidity on stock exchanges.
- Pricing can be at nominal, premium, or discount, as per rules.
- Can be enhanced, reduced, split, bought back, or issued free, per regulations.
- Shares can be pledged for loans, as they are marketable securities.

Other Important Term:

Authorised Capital / Nominal Capital:

- ✓ The maximum capital a company can issue.
- ✓ Decided by the Board and stated in the Memorandum & Articles of Association.
- ✓ Changes require shareholder approval as per the Companies Act.

Equity Capital

❖ Authorised/Nominal Capital

- ❖ Maximum capital a company can issue.
- ❖ Specified in the Memorandum & Articles of Association.
- ❖ Changes require shareholder approval as per the Companies Act.

❖ Issued Share Capital

- ❖ Portion of authorised capital offered to public for subscription.
- ❖ Includes shares issued for cash or other considerations.

❖ Subscribed Share Capital

- ❖ Portion of issued capital subscribed by the public.
- ❖ May include shares issued for non-cash consideration.
- ❖ Actual subscription may be less than the called-up capital.



5CV

❖ Called-up Share Capital

- ❖ Portion of subscribed capital that the company calls upon shareholders to pay.
- ❖ Can be partial or full, depending on need and regulations.

❖ Paid-up Share Capital

3CV

- ❖ Actual amount received from shareholders against called-up capital.
- ❖ May be less than subscribed capital if some payments are pending.

❖ Calls in Arrears

2CV

- ❖ Unpaid part of called-up capital.
- ❖ The company can recover it from defaulting shareholders.

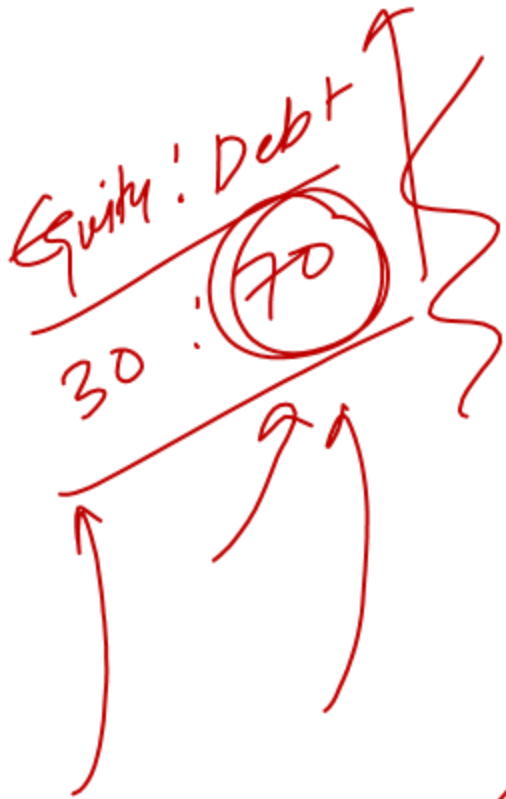
❖ Unpaid Share Capital

1CV

- ❖ Final amount determined as unpaid by the management.

❖ Forfeited Shares

- ❖ Shares not fully paid for, and forfeited by the company.
- ❖ Can be reissued as per board decision and regulations.

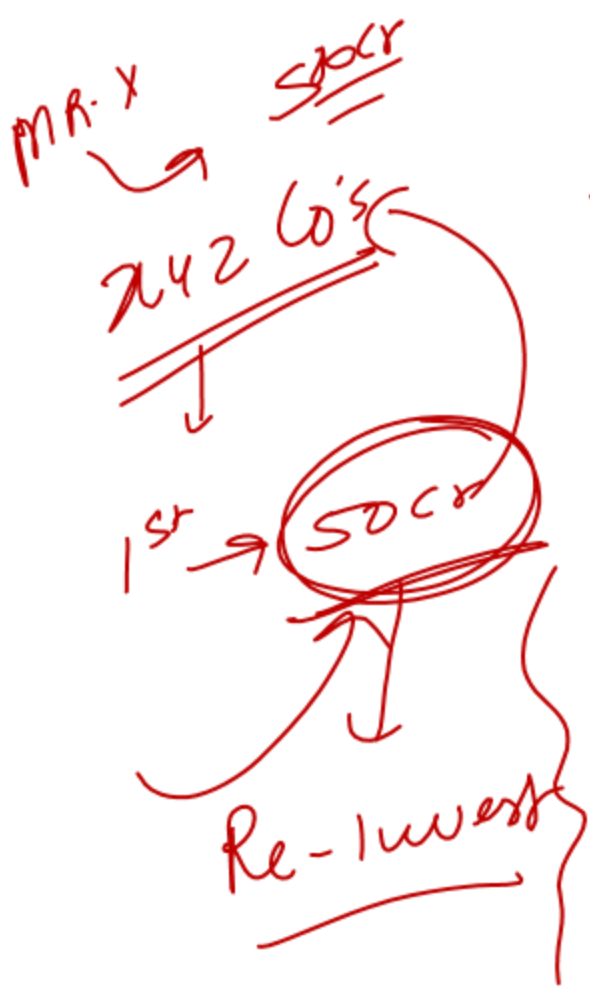


Reasons for Choosing Equity over Loans:

- ✓ No mandatory reward or interest to be paid to equity holders.
- ✓ No repayment obligation, unlike loans.
- ✓ Provides risk capital that is otherwise hard to obtain.
- ✓ Offers ownership to the equity holder.
- ✓ Higher equity boosts lender confidence and project safety.

Drawbacks of Equity Capital:

- ❖ High return expectations from equity shareholders due to higher risk borne by them — leading to higher cost of capital.
- ❖ Cost of raising equity is high, including:
 - ✓ Advertising
 - ✓ Printing
 - ✓ Brokerage
 - ✓ Commissions (brokerage, underwriting)
 - ✓ Compliance costsThese make equity more expensive compared to other funding sources.



Internal Accruals

Internal Accruals refer to the retained portion of Profit After Tax (PAT) that is not distributed as dividends but kept within the company. These are used as a prudent financial source and are reflected in reserves.

Sources of Internal Accruals:

- ✓ Profit After Tax (PAT)
- ✓ Non-cash profits like provisions/reserves
- ✓ Depreciation

These funds are available for company use even after paying dividends on preference shares and may be used for issuing bonus shares or transferred to free reserves after fulfilling legal requirements.

W.W.

Advantages:

- ✓ No restriction on usage (except statutory limitations).
- ✓ Suitable for both long-term and short-term use.
- ✓ No cost for raising or servicing these funds.
- ✓ Readily and easily available.
- ✓ As liquid as the form they are invested in.
- ✓ No dilution of ownership/control.
- ✓ No fund-raising cost involved.

Disadvantages:

- ✓ Risk of **indiscriminate usage** due to ease of access.
- ✓ **Higher cost in reality** as they represent **undistributed dividends**, and equity holders expect returns.
- ✓ Retained earnings **reduce dividends**, affecting especially **minority shareholders**.
- ✓ Excessive retention can hurt the company's **dividend track record**, reducing public investor interest.

Preference Capital

Preference Capital refers to the capital raised by issuing preference shares, which offer investors **priority over equity shareholders** in receiving dividends and during liquidation, though not as secure as debt.

Key Features:

- ✓ **Priority in dividend and liquidation** over equity shares.
- ✓ **Fixed dividend rate**, like debentures.
- ✓ Dividend is **paid only if distributable profits exist**, and not mandatory.
- ✓ Dividend is **paid from post-tax profits** and is **not tax-deductible**.
- ✓ **Cumulative preference shares** accumulate unpaid dividends for future payment.
- ✓ **Redeemable preference shares** must be repaid within 20 years, from profits or fresh issue.
- ✓ **Voting rights are limited**, except when dividends are unpaid for 2+ years.

Reasons for Raising Preference Capital:

- ✓ Suitable for **long-term funding** (up to 20 years or more).
- ✓ **Cost-effective**, especially when **privately placed**.
- ✓ No compulsory dividend unless **cumulative preference shares** are issued.
- ✓ **Predictable cost**, as dividend is fixed and known in advance.
- ✓ **No dilution of equity**, allowing promoters to **retain control**.

✓ 7.5 TERM LOANS

g.w.

This is another major source of long-term finance by which a company can obtain term loans from banks or financial institutions. While banks also give working capital finance, the Financial Institutions are allowed to give only such loans which are repayable over the years as per the terms and conditions of the sanction. This source is different from equity and relatively, cheaper to service. The equity forms are either a permanent source where repayment is not there or a very long period like 20 years is available for repayment in the case of preference shares or capital. Term loans can be for long to medium terms stretching to about ten years. Term loan may not be as costly a source of finance as taking working capital finance because rates are determined based on medium to long term prospects of the project where cyclical movements in the business fortune are more or less evened out. Generally, term loans are resorted to finance setting up or expansion of the projects.

...and it is mandatory to agree for the memorandum for the interest payments.

A list of typical terms, conditions and requirements incorporated in the loan agreements.

- Reconstitution of the board.
- Induction of the independent directors.
- A seat on the board.
- Prior consent or approval of the lender required for following actions:
Examples; Mergers, hiving off, restructuring, new projects, equity expansion or dilution, investments in or creation of subsidiaries, fresh or big funding exercise etc.
- Statutory registrations or approvals.
- Infusion of additional funds by the promoters.
- Prohibition on withdrawal of loans or funds already brought in by the promoters.
- Consent required of the lender if any other loans are being repaid.
- Restrictions on dividend pay-outs.
- Submission of quarterly data and annual audited financial statements.
- Requirements of rating by independent and reputed rating agency.
- Ceiling on further borrowings.
- Inspections and visits.
- Insisting on First charge failing which pari-passu charge.
- Prohibition on creating further charges.
- Pledge of promoters' shares or collaterals of personal assets of the promoters.
- Restrictions on promoters' right to dispose their shares.
- Appointment of compliance officers and proper Key Managerial Persons.
- Strong Corporate governance.
- Maintenance of prudent financial ratios including debt equity.

Debentures

Definition:

As per Section 2(30) of the Companies Act, 2013, a *debenture* is a written instrument acknowledging a company's debt. It includes debenture stock, bonds, or any other debt instrument, with or without a charge on the company's assets.

Key Features of Debentures:

✓ Instrument of Debt:

- ✓ Debentures are promises by companies to repay a fixed amount on a future date.

✓ Legal Deed:

- ✓ For issues involving over 500 persons, trustees must be appointed to protect debenture holders' interests.

✓ Regulatory Framework:

- ✓ Section 71 of the Companies Act, 2013 outlines provisions for issuance, voting rights, redemption, convertibility, etc.



- ✓ **Fixed Tenure:** Repayment is done either in instalments or on a fixed maturity date (redemption). Perpetual bonds can also be issued.
- ✓ **Fixed Interest Rate:** A pre-decided interest rate is applicable. For *Zero-Coupon Bonds*, interest is adjusted via discounting on redemption.
- ✓ **Issuer Options (Call Option):** Company can repay before due date at a predetermined price.
- ✓ **Investor Options (Put Option):** Investor may demand early redemption before due date.
- ✓ **No Voting Rights:** Section 71(2) prohibits voting rights for debenture holders.
- ✓ **Convertible Debentures:** Convertible (fully/partly) debentures can be turned into equity, giving holders equity stake and voting rights as per agreed terms.

is indicated in the offer document.

Types of debentures.

- **Based on tenure:** There are debentures specifying redemption with call options, put options, or with fixed tenure under this kind. Perpetual bonds with call option, are also permitted to be issued.
- **Based on security:** Many a times, to offer lower rate of interest and a sense of security to the investors, secured debentures are issued providing first or second charge over the fixed assets of the company and appointing trustees if the number of holders exceeds 500.
- **Based on Convertibility:** To provide additional incentive to the investors, the companies issue partly or fully convertible debentures so that at a future date, upon conversion, the debentures holders can be growth participants and also have voting rights.
- **Based on negotiability:** The debenture is a debt instruments and mostly transferable by way of a registered transfer form. However, in case of bearer debentures the transfer takes place by mere delivery. Such debentures are rare due to concerns such as money laundering and benami transactions.

Alternative Financing Strategies in the Context of Regulatory Requirements

Overview:

- ✓ Financing is vital for any business but involves risks and must be regulated.
- ✓ Traditional finance is regulated mainly by the RBI under the **Banking Regulation Act, 1949**, and applies to banks and NBFCs.
- ✓ RBI also issues norms for **non-banking finance**, which are increasingly important in the corporate sector.
- ✓ **SEBI** regulates many corporate-sector financing options and proposed (but didn't pursue) regulations for **crowd funding**.

Alternative Financing Options:

These cater especially to **SMEs, start-ups, and sectors like real estate** and include:

- ✓ **Crowd funding**
- ✓ **Peer-to-peer lending**
- ✓ **Alternate finance funds**
- ✓ **Mutual funds**
- ✓ **Angel investors**
- ✓ **Venture capital funds**
- ✓ **Bootstrapping**
- ✓ **REITs (Real Estate Investment Trusts)**
- ✓ **InvITs (Infrastructure Investment Trusts)**

These offer **equity or debt financing** from **micro to extra-long term** but fall **outside the purview of banks/NBFCs**.

Private Equity (PE)

Definition & Nature:

- ✓ PE provides medium to long-term equity capital to **non-listed companies** with **high growth potential**.
- ✓ Known as “**patient capital**”, it seeks **long-term capital gains** rather than quick returns.
- ✓ PE is **illiquid** and suited for **long-term investments**.

Forms of Private Equity Investment:

- ✓ **Venture Capital**
- ✓ **Leveraged Buyouts (LBOs)**
- ✓ **Mezzanine Investments**
- ✓ **Distressed Debt**
- ✓ **Build-ups**

Stages of PE Financing (per Pratt, 1981):

- ✓ **Seed Financing** – Initial capital to develop a business idea.
- ✓ **Start-up Financing** – Funds for product development and early marketing.
- ✓ **First-stage Financing** – For commercialization and production.
- ✓ **Second-stage Financing** – Working capital for growth-phase companies.



7.7.2 Foreign Direct Investment (FDI)

Foreign Investment in India is governed by the FDI policy announced by the Government of India and the provisions of the Foreign Exchange Management Act (FEMA) 1999. Foreign investment is freely permitted in almost all sectors. Foreign Direct Investments (FDI) can be made under two routes—Automatic Route and Government Route. Under the Automatic Route, the foreign investor or the Indian company does not require any approval from RBI or Government of India for the investment. Under the Government Route, prior approval of the Government of India, Ministry of Finance, Foreign Investment Promotion Board (FIPB) is required. Indian companies can freely issue equity shares / convertible debentures and preference shares subject to valuation norms prescribed under FEMA Regulations. Issue of other types of preference shares such as non-convertible, optionally convertible or partially convertible are considered as debt. As such, the guidelines applicable for External Commercial Borrowing (ECB), viz. eligible borrowers, recognised lenders, amount and maturity, end use stipulations, etc. will apply to such issues. A foreign investor can invest in an Indian company which is a small scale industrial unit provided it is not engaged in any activity which is prohibited under the FDI policy. Such investments are subject to a limit of 24% of paid-up capital of the Indian company/SSI Unit.

The policy and procedures in respect of FDI in India is available in 'the Manual on Investing in India- Foreign Direct Investment, Policy & Procedures' which is available in the public domain.



FINANCIAL AND OPERATING LEVERAGES

STRUCTURE

- 8.0 Objectives
- 8.1 Introduction
- 8.2 Financial Leverage
- 8.3 Degree of Financial Leverage and Its Behaviour
- 8.4 Operating Leverage
- 8.5 Degree of Operating Leverage and Its Behaviour
- 8.6 Combined or Total Leverage

Let Us Sum Up

Keywords

In financial analysis, **leverage** reflects the relationship between fixed costs and profitability, where leverage magnifies the impact of sales fluctuations on a business's earnings. There are two primary types of leverage:

Financial Leverage: Involves **fixed financial costs** (mainly interest expenses on debt). Correlates debt obligations to profits, using **borrowed funds to increase returns on equity** as long as the return from assets purchased with debt exceeds the borrowing cost.

Leverage involves using **borrowed funds (debt)** to enhance EPS without additional equity capital. Borrowed funds are used to acquire assets, which generate value exceeding the cost of debt, increasing EPS for shareholders.

Leverage Concept

- ✓ **Financial leverage** shows how sensitive owners' cash flows are to changes in operating cash flows.
- ✓ **Operating leverage** indicates how operating cash flows respond to changes in sales.
- ✓ Businesses must assess **both types of leverage** because they affect owners' future cash flow risks.
- ✓ Managers must align investment and financing decisions to maximize owners' wealth, balancing risk and return.

Degree of Total Leverage (DTL):

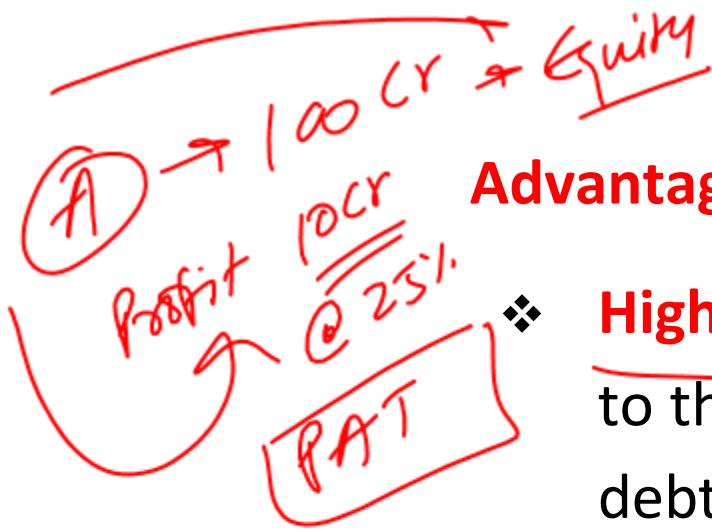
- ✓ Measures how sensitive **owners' cash flows** are to changes in **unit sales**.
- ✓ Combines **operating leverage** and **financial leverage**.





Earnings Per Share (EPS): EPS is the primary measure of return for equity shareholders. Without debt, EPS is driven solely by the business's net profits divided by the number of equity shares.

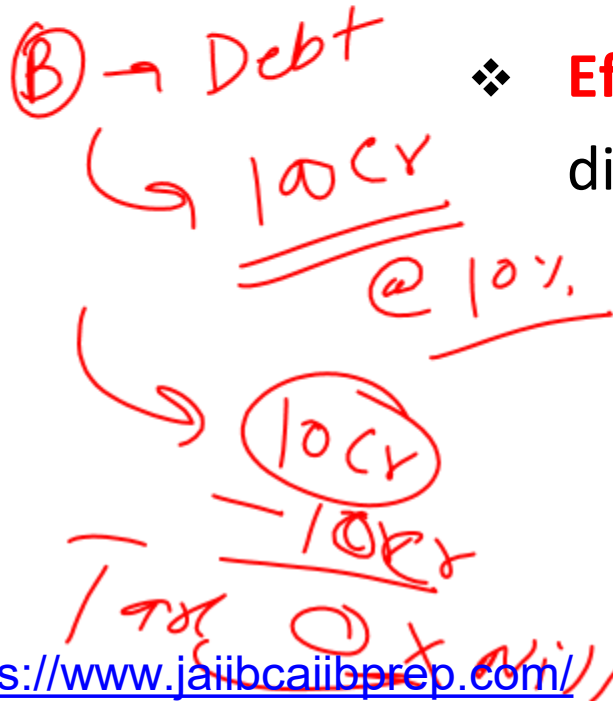
Impact of Equity Financing: Increasing equity capital boosts absolute profit but keeps EPS unchanged (as profits are spread over more shares).



Advantages of Financial Leverage:

~~PAT - Div~~

- ❖ **Higher EPS:** Borrowed funds lead to asset acquisition, adding value to the business if the return from these assets exceeds the cost of debt.



- ❖ **Efficient Capital Usage:** Owners can increase returns without diluting ownership by issuing additional shares.

Constraints on Financial Leverage:

1. **Increased Debt Ratio:** As debt-to-equity ratio rises, lenders become cautious, leading to: Refusal to lend more. Higher interest rates, which may erode the profitability of additional borrowing.
2. **Business Risk:** High debt reduces the business's ability to cushion adverse conditions, potentially: Lowering EPS if returns on assets are less than borrowing costs. Increasing financial instability during downturns.

Prudential Borrowing Limits:

Owners typically cap borrowings to ensure:

- ❖ Financial stability.
- ❖ Maintenance of acceptable debt servicing costs.
- ❖ Sustainable growth without excessive risk exposure.

Conclusion:

Financial leverage can be a powerful tool to enhance shareholder returns, but its benefits are subject to careful management of debt levels and associated risks. Optimal leverage is achieved when borrowed funds add value exceeding their cost, without overburdening the business financially.



We can write it as:

$$\text{Degree of Financial Leverage (DFL)} = \frac{\text{Earnings before interest and tax (EBIT)}}{\text{Earnings before tax (EBT)}}$$

Where, $\text{EBIT} = \text{Sales} - (\text{Variable cost} + \text{Fixed cost excluding interest})$

$$\text{EBT} = \text{EBIT} - \text{Interest}$$

ILLUSTRATION 1:

A firm's details are as under:

Sales (@100 per unit) Rs. 24,00,000

Variable Cost 50%

Fixed Cost Rs. 10,00,000

It has borrowed Rs. 10,00,000 @ 10% p.a. and its equity share capital is Rs. 10,00,000 (Rs. 100 each).

Consider tax @ 50 %.

Calculate its Degree of Financial Leverage

2L

12L

1 = 14

$$\frac{2L}{14} = 2$$

SOLUTION:

	(Rs.)
Sales	24,00,000
Less: Variable cost	12,00,000
Contribution	12,00,000
Less: Fixed cost	10,00,000
EBIT	2,00,000
Less: Interest	1,00,000
EBT	1,00,000
Less: Tax (50%)	50,000
EAT	50,000
No. of equity shares	10,000
EPS	5

$$\text{Degree of Financial Leverage} = \frac{\text{EBIT}}{\text{EBT}} = \frac{2,00,000}{1,00,000} = 2 \text{ times}$$

ILLUSTRATION 2

The following information is related to XYZ Company Ltd. for the year ended 31st March, 2022:

Equity share capital (of Rs. 10 each) Rs. 50 lakhs

12% Bonds of Rs. 1,000 each Rs. 37 lakhs @ 12%

Sales Rs. 84 lakhs

Fixed cost (excluding interest) Rs. 6.96 lakhs

Profit-volume Ratio 27.55%

Income Tax Applicable 40%

Calculate its Degree of Financial Leverage

$$FL = \frac{EBIT}{EBT} = \frac{16,18,200}{11,74,200} = 1.38$$

$$C = SP - VC$$

SOLUTION:

Particulars	Amount (Rs.)
Sales	84,00,000
Contribution (Sales $\times$ P/V ratio)	23,14,200 ✓
Less: Fixed cost (excluding Interest)	✓ (6,96,000)
EBIT (Earnings before interest and taxes)	16,18,200
Less: <u>Interest on debentures</u> (12% Rs.37 lakhs)	(4,44,000)
EBT (Earnings before tax)	11,74,200

$$\text{Degree of Financial Leverage} = \frac{\text{EBIT}}{\text{EBT}} = \frac{16,18,200}{11,74,200} = 1.38 \text{ times}$$

Operating Leverage: Focuses on **fixed operational costs**, excluding interest, and their relationship to profit. Measures how sensitive a company's earnings are to changes in sales volume due to high fixed costs in operations.

Examples of Operating Leverage:

Low Operating Leverage (e.g., Wholesaler): Low fixed costs and low contribution margin (sales price - purchase price). Less sensitive to sales volume changes, as low fixed costs mean profits remain stable even if sales decrease.

High Operating Leverage (e.g., Hospital): High fixed costs (e.g., facilities, equipment) with relatively low variable costs. Profits are more sensitive to fluctuations in sales volume; a decline in business can severely impact earnings.

A business with **high operating leverage** experiences significant fluctuations in **Earnings Before Interest and Taxes (EBIT)** as sales volume changes, leading to potential volatility in return on invested capital.

Degree of Operating Leverage (DOL):

- DOL quantifies a business's sensitivity to changes in sales volume and the resulting effect on EBIT.
- Formula:
$$DOL = \frac{\% \text{ Change in EBIT}}{\% \text{ Change in Sales Volume}}$$
- Higher DOL indicates greater sensitivity to sales changes, with a larger impact on profits.



$$DOL = \frac{\text{Contribution}}{\text{Earnings Before Interest and Taxes (EBIT)}}$$

Illustration 1:

Calculate Degree of Operating Leverage when, a company sells 1000 units of product X at Rs. 50 having variable cost of Rs. 30 per unit and fixed cost of Rs. 15,000.

Solution:

Particulars	Product X (Rs.)
Sales (50 x 1000 units)	50,000
Variable Cost (30 x 1000 units)	<u>30,000</u>
Contribution	<u>20,000</u>
Fixed Cost	<u>15,000</u>
Profit (EBIT)	<u>5,000</u>

$$BEP = \frac{FC}{C}$$


$$\text{Degree of Operating Leverage} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{20,000}{5,000} = 4 \text{ times}$$

Illustration 2:

A Company ABC Ltd produces and sells 20,000 shirts. The selling price per shirt is Rs. 500. Variable cost is Rs. 200 per shirt and fixed operating cost is Rs. 40,00,000. Calculate degree of operating leverage.

Particulars	Rs.
Sales Revenue (20,000 × 500)	1,00,00,000
Less: Variable Cost (20,000 × 200)	40,00,000
Contribution	60,00,000
Less: Fixed Cost	40,00,000
EBIT	20,00,000

$$\text{Degree of Operating Leverage} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{60,00,000}{20,00,000} = 3 \text{ times}$$

Illustration 3:

Calculate the degree of operating leverage for each of the four Companies A Ltd, B Ltd, C Ltd and D Ltd from the following price and cost data:

	Company			
	A Ltd (Rs.)	B Ltd (Rs.)	C Ltd (Rs.)	D Ltd (Rs.)
Sale price per unit	20	32	50	70
Variable cost per unit	6	16	20	50
Fixed operating cost	60,000	40,000	1,00,000	Nil

Solution:

	Company			
	A Ltd (Rs.)	B Ltd (Rs.)	C Ltd (Rs.)	D Ltd (Rs.)
Sales (units)	5,000	5,000	5,000	5,000
Sales revenue (Units × sale price per unit)	1,00,000	1,60,000	2,50,000	3,50,000
Less: Variable cost (Units × variable cost per unit)	(30,000)	(80,000)	(1,00,000)	(2,50,000)
Contribution	70,000	80,000	1,50,000	1,00,000
Less: Fixed operating costs	(60,000)	(40,000)	(1,00,000)	Nil
EBIT	10,000	40,000	50,000	1,00,000

$$\text{Degree of Operating Leverage A Ltd} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{70,000}{10,000} = 7 \text{ times}$$

$$\text{Degree of Operating Leverage B Ltd} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{80,000}{40,000} = 2 \text{ times}$$

$$\text{Degree of Operating Leverage C Ltd} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{1,50,000}{50,000} = 3 \text{ times}$$

$$\text{Degree of Operating Leverage D Ltd} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{1,00,000}{1,00,000} = 1 \text{ times}$$

When DOL is more than one (1), operating leverage is relevant. A positive DOL/OL means that the firm is operating at higher level than the break-even level and both sales and EBIT moves in the same direction. In case of negative DOL/OL, firm operates at lower than the break-even sales and EBIT is negative.

8.6 COMBINED OR TOTAL LEVERAGE

This is a combination of financial and operative leverages. Here the fixed costs include both the operating fixed costs and financial fixed costs (interest). In such a situation, a marginal percentage change in revenue leads to disproportionate percentage change in EBT. For example, a 10% increase in revenue may increase EBT and consequently, the EPS by 15%. Similarly, a 10% fall in revenue may decrease EBT and consequently, the EPS by more than 10%. The following formula is used for calculating the degree of total leverage (DTL)

$$\text{Degree of Total leverage (DTL)} = \frac{\text{Percentage change in EBT}}{\text{Percentage change in Sales}}$$

It is a product of both financial and operating leverages and can be written as:

$$\text{DTL} = \text{DFL} \times \text{DOL}$$

$$= \text{EBIT/EBT} \times \text{Contribution/ EBIT}$$

$$= \text{Contribution/EBT}$$

ILLUSTRATION 4

A firm has sales of Rs. 10,00,000, variable cost of Rs. 7,00,000 and fixed costs of Rs. 2,00,000 and debt of Rs. 5,00,000 at 10% rate of interest. What are the operating, financial and combined leverages?

SOLUTION:

Statement of Profit		(Amt in Rs.)
Sales		10,00,000
Less: Variable Cost		7,00,000
Contribution	$PL = \frac{1L}{50K} 2$	3,00,000
Less: Fixed Cost	$OL = \frac{3L}{1L} 3$	2,00,000
EBIT		1,00,000
Less: Interest @ 10% on 5,00,000	$OL \times FL = 6$	50,000
Earnings before tax (EBT)		50,000

$$\text{Operating Leverage} = \frac{\text{Contribution}}{\text{EBIT}} = \frac{3,00,000}{1,00,000} = 3$$

$$\text{Financial Leverage} = \frac{\text{EBIT}}{\text{EBT}} = \frac{1,00,000}{50,000} = 2$$

$$\text{Combined Leverage} = \frac{\text{Contribution}}{\text{EBT}} = \frac{3,00,000}{50,000} = 6$$

ILLUSTRATION 5

Z Ltd has estimated that for a new product, the break-even point is 2,000 units, if the items are sold for Rs14 per unit. The Cost Accounting department has currently identified variable cost of Rs. 9 per unit. Calculate the degree of operating leverage for sales volume of 2,500 units and 3,000 units. What do you infer from the degree of operating leverage at the sales volumes of 2,500 units and 3,000 units and their difference if any?

SOLUTION:

Statement of Operating Leverage

Particulars	2,500 Units	3,000 Units
Sales @ Rs. 14 per unit	35,000	42,000
Variable cost @ Rs. 9 per unit	22,500	27,000
Contribution	12,500	15,000
Fixed cost Rs. [2,000 × (14 - 9)]	10,000	10,000
EBIT	2,500	5,000
Degree of Operating Leverage = $\frac{\text{Contribution}}{\text{EBIT}}$	$\frac{12,000}{2,500}$	$\frac{15,000}{5,000}$
Degree of Operating Leverage	5	3

UNIT 9

CAPITAL INVESTMENT DECISIONS

STRUCTURE

- 9.0 Objectives
- 9.1 Introduction
- 9.2 Objective of Capital Investment Decisions
- 9.3 Estimation of Project Cash Flows
- 9.4 Social Cost Benefit Analysis

Let Us Sum Up



Estimation of Project Cash Flows

Cash flow estimation involves assessing:

- ✓ Long-term vs. Short-term needs.
- ✓ Owned vs. Borrowed capital.
- ✓ Fixed costs vs. Working capital.
- ✓ Inflows vs. Outflows.



Elements of Cash Flow:

1. **Initial:**

Includes capital expenditure and working capital to start the project.

2. **Operating:**

Covers operational inflows and outflows (e.g., sales revenue, production costs).

3. **Terminal:**

Net inflows after liquidating assets at the project's end.

Principles of Cash Flow Estimation:

1. **Separation Principle:** Differentiate between investment-related and financing-related cash flows.
2. **Incremental Principle:** Separate costs incurred regardless of project execution (fixed costs) from project-specific costs (variable costs).
3. **Post-Tax Principle:** Use post-tax cash flows to ensure reliability of discount rates.
4. **Consistency Principle:** Maintain consistency across all estimations.

Statement showing the calculation of Cash Inflow After Tax (CFAT)

Particulars	(Rs.)	(Rs.)
Sales value		XXX
Less: Variable Cost		XXX
Contribution		XXX
Less: Fixed Cost		
(a) Fixed Cash Cost	XXX	
(b) Depreciation	XXX	XXX
Earning Before Tax (EBT)		XXX
Less: Tax		XXX
Earning After Tax (EAT)		XXX
Add: Depreciation		XXX
Cash Inflow After Tax (CFAT)		XXX

Q. XYZ Ltd is evaluating the purchase of a new machinery with a depreciable base of Rs. 1,00,000; expected economic life of 4 years and change in earnings before taxes and depreciation of Rs. 45,000 in 2021, Rs. 30,000 in year 2022, Rs. 25,000 in year 2023 and Rs. 35,000 in year 2024. Assume straight-line depreciation and a 20% tax rate. You are required to compute relevant cash flows.

SOLUTION:

Depreciation = 1,00,000 divided by 4 = Rs. 25,000 per year

Amount in (Rs.)

	Years			
	2021	2022	2023	2024
Earnings before tax and depreciation	45,000	30,000	25,000	35,000
Less: Depreciation	(25,000)	(25,000)	(25,000)	(25,000)
Earnings before tax	20,000	5,000	0	10,000
Less: Tax @20%	(4,000)	(1,000)	0	(2,000)
Earnings after tax	16,000	4,000	0	8,000
Add: Depreciation	25,000	25,000	25,000	25,000
Net Cash flow	41,000	29,000	25,000	33,000

✓ 10L PV of Future Inflow

Net Present Value (NPV)

Net Present Value (NPV) is a financial metric used to evaluate the profitability of an investment. It represents the difference between the present value (PV) of future cash inflows and the initial investment (outflows). NPV accounts for the time value of money, emphasizing that money today is worth more than the same amount in the future.

✓ R = 10%.

$$\frac{\text{Inflow}}{(1+r)^1} + \frac{\text{Inflow}}{(1+r)^2} + \frac{\text{Inflow}}{(1+r)^3} = \text{lowest.}$$

$$\frac{2L}{(1+0.1)^1} + \frac{4L}{(1+0.1)^2} + \frac{6L}{(1+0.1)^3} - 10L = \boxed{NPV > 0} \checkmark$$

Formula

$$\text{NPV} = \sum \left(\frac{\text{Net Cash Inflow}_t}{(1 + r)^t} \right) - \text{Initial Investment}$$

Where:

- t : Time period (year 1, 2, ... n).
- r : Discount rate (expected rate of return).
- Net Cash Inflow_t : Net cash flow in period t .

Payback Period and Discounted Payback Period

Payback Period

The time it takes for the initial investment in a project to be fully recovered through net cash flows.

- Key Formula (Uniform Cash Inflows):

$$\begin{array}{|l} \hline (100) \\ \hline \end{array} \quad \begin{array}{|l} 1 \rightarrow 3L \\ 2 \rightarrow 3L \\ 3 \rightarrow 4L \\ \hline \end{array}$$

$$\text{Payback Period} = \frac{\text{Initial Investment}}{\text{Annual Net Cash Inflow}}$$

= 3.48

T.W.

The following points are worth noting:

- This method simply identifies the period in which the initial investment in the project is fully recovered.
- It does not take time value of money into account unless it is discounted payback period method.
- The method is a simple concept.
- This is suitable for small businesses where it is not worth spending more time and money to make more detailed or scientific analysis.
- The shorter the period for recovery of initial investment, lesser is the risk and better or quicker is the probability to earn profits.
- Payback Period method ignores the cash flow beyond the payback period and hence ignores profit aspect.
- While payback period method is used across the line, the other more sophisticated methods are used in larger sized budgeting exercise.

Consider, a firm wants to set up a toy making plant. Production of the toys requires new factory space and equipment. Either the firm can construct a factory on a site it has identified or it can refit a factory that it owns and that is currently lying idle. The initial investment in constructing a factory is more; however, the running cost may be less due to a design that is more appropriate.

The initial investment in renovating the old factory is comparatively low; however, the running cost may be high due to refit of the factory.

The firm is faced with a choice of mutually exclusive investment opportunities. The net cost associated with the newly built factory (Project A) and that of the renovated factory (Project B) are shown in Tables 25.1 and 25.2 respectively, as are the present values of the cash flows for a discount rate of 10 per cent.

TABLE 25.1 : FOR PROJECT A

Year	Cash Inflow ('000)	Discounting Factor (@10%)	Present Value ('000)
0	- 2000	1	- 2000
1	350	0.909	318.18
2	500	0.826	413.22
3	600	0.751	450.79
4	800	0.683	546.41
5	800	0.621	496.74
6	600	0.564	338.68
		NPV	+564.02

TABLE 25.2 : FOR PROJECT B

Year	Cash Inflow ('000)	Discounting Factor (@10%)	Present Value ('000)
0	-1500	1	-1500
1	200	0.909	181.82
2	260	0.826	214.88
3	450	0.751	338.09
4	700	0.683	478.11
5	700	0.621	434.64
6	400	0.564	225.79
		NPV	+373.33

Out (1L)	PV of Future Inflows
(PV)	
1 → 30K	
2 → 30K	
3 → 30K	
4 → 30K	
5 → 30K	

Q. ABC Company is considering an investment project that requires an initial investment of Rs.1,00,000. The project is expected to generate cash inflows of Rs. 30,000 per year for the next 5 years. The discount rate used for evaluating the project is 10%.

What is the NPV of the project? Should the project be undertaken?

$$\frac{30K}{(1+0.10)^1} - \dots - 1L = \boxed{}$$

(A) 10,722, Yes

(B) - 10,722, No

(C) 13,722, Yes

(D) - 13,722, No

Cash Flows	Present Value
Year 1: 30,000	$\frac{30,000}{(1+0.10)^1} = 27,273$
Year 2: 30,000	$\frac{30,000}{(1+0.10)^2} = 24,793$
Year 3: 30,000	$\frac{30,000}{(1+0.10)^3} = 22,539$
Year 4: 30,000	$\frac{30,000}{(1+0.10)^4} = 20,490$
Year 5: 30,000	$\frac{30,000}{(1+0.10)^5} = 18,627$

$$\text{NPV} = 27,273 + 24,793 + 22,539 + 20,490 + 18,627 - 1,00,000 = 113,722 - 1,00,000 = 13,722$$

Since the NPV is positive, the project should be undertaken.

-(14)

fv of 1 ✓

- ① $30k \times .909 = \boxed{}$
+
② $30k \times .826 = \boxed{}$
③ $30k \times .751 = \boxed{}$
④ $30k \times .683 = \boxed{}$
⑤ $30k \times .621 = \boxed{}$

Q. Company XYZ is considering investing in a new project that requires an initial investment of Rs. 1,00,000. The project is expected to generate cash flows of Rs. 30,000 per year for the next five years. The company's required rate of return for similar projects is 10%. Determine the NPV of the project using the given table of discounting factor.

Year	1	2	3	4	5
Discounting Factor @ 10%	0.909	0.826	0.751	0.683	0.621

- (1) 12,700
(3) 14,700

- (2) 13,700
(4) 15,700

+ ✓

Solution (2)

	Cash inflow/outflow	Discounting Factor	Present Value
Year 0: Cash outflow	-1,00,000	1	-1,00,000
Year 1: Cash inflow	30,000	0.909	27,270
Year 2: Cash inflow	30,000	0.826	24,780
Year 3: Cash inflow	30,000	0.751	22,530
Year 4: Cash inflow	30,000	0.683	20,490
Year 5: Cash inflow	30,000	0.621	18,630
		NPV	13,700

Since the NPV is positive, the project should be undertaken.



XYZ Ltd wants to expand its business and so it is willing to invest Rs 10,00,000. The investment is said to bring an inflow of Rs. 1,00,000 in first year, 2,50,000 in the second year, 3,50,000 in third year, 2,65,000 in fourth year and 4,15,000 in fifth year. Assuming the discount rate to be 9%. Let us calculate NPV using the formula.

$$NPV = \frac{Cash\ Flow_1}{(1+r)^1} + \frac{Cash\ Flow_2}{(1+r)^2} + \frac{Cash\ Flow_n}{(1+r)^n} - Initial\ Investment$$

Accounting Rate of Return (ARR): The **ARR** is a method of evaluating an investment's profitability by comparing the average accounting profit to the initial investment or average investment. Unlike other methods, ARR focuses on accounting data rather than cash flow and is widely used due to its simplicity and accessibility.

Formula

$$ARR = \frac{\text{Average Annual Accounting Profit}}{\text{Initial Investment}} \times 100$$

Alternatively, if the average investment is considered:

$$ARR = \frac{\text{Average Annual Accounting Profit}}{\text{Average Investment}} \times 100$$

Where:

- $\text{Average Investment} = \frac{\text{Initial Investment} + \text{Salvage Value}}{2}$.

ILLUSTRATION 6

A project, requiring an investment of Rs. 10,00,000, yields profit after tax and depreciation which is as follows:

Years	Profit after tax and depreciation (Rs.)
1	50,000
2	75,000
3	1,25,000
4	1,30,000
5	80,000
Total	4,60,000

Suppose further that at the end of the 5th year, the plant and machinery of the project can be sold for Rs. 80,000. Determine the Accounting Rate of Return.

SOLUTION:

In this case the rate of return can be calculated as follows:

$$\frac{\text{Total Profit} \div \text{No. of years}}{\text{Average investment or Initial Investment}} \times 100$$

(a) If Initial Investment is considered then,

$$= \frac{4,60,000 \div 5 \text{ years}}{10,00,000} \times 100 = 9.2\%$$

(b) If Average investment is considered, then,

$$= \frac{\text{Rs. 92,000}}{\text{Avg Investment}} \times 100 = \frac{\text{Rs. 92,000}}{\text{Rs. 5,40,000}} \times 100 = 17.04\%$$

Where,

$$\begin{aligned} \text{Average Investment} &= \frac{1}{2} (\text{Initial investment} - \text{Salvage value}) + \text{Salvage value} \\ &= \frac{1}{2} (\text{Rs. 10,00,000} - \text{Rs. 80,000}) + \text{Rs. 80,000} \\ &= \text{Rs. 4,60,000} + \text{Rs. 80,000} = \text{Rs. 5,40,000} \end{aligned}$$

Investment Proposals:

- **Proposal X:**

Initial Investment = ₹2,00,000

Life = 4 years

Estimated annual profits = ₹20,000, ₹40,000, ₹50,000, ₹30,000.

- **Proposal Y:**

Initial Investment = ₹2,00,000

Life = 4 years

Estimated annual profits = ₹25,000, ₹35,000, ₹45,000, ₹15,000.

Step 1: Calculate the **average annual profit** for each proposal:

$$\text{Average Annual Profit (X)} = \frac{20,000 + 40,000 + 50,000 + 30,000}{4} = ₹35,000$$

$$\text{Average Annual Profit (Y)} = \frac{25,000 + 35,000 + 45,000 + 15,000}{4} = ₹30,000$$

Step 2: Calculate ARR:

For **Proposal X**:

$$\text{ARR} = \frac{35,000}{2,00,000} \times 100 = 17.5\%$$

For **Proposal Y**:

$$\text{ARR} = \frac{30,000}{2,00,000} \times 100 = 15\%$$

9.5.3.6 Profitability Index

This is an index that either explains or represents the relationship between the cost and the benefit of a project proposal. It is also called value investment ratio or profit investment ratio. PI is calculated by dividing the present value of future expected cash flows by the initial investment amount in the project. Higher the Index better is profitability of the project. Anything below 1 indicates that the project is unprofitable.

Mathematically: The Profitability Index (PI) is calculated as below:

$$\text{Profitability Index (PI)} = \frac{\text{Sum of discounted cash inflows}}{\text{Initial cash outlay or Total discounted cash outflow (as the case may be)}}$$

ILLUSTRATION 11

Suppose we have three projects involving discounted cash outflow of Rs. 5,50,000, Rs. 75,000 and Rs. 1,00,20,000 respectively. Suppose further that the sum of discounted cash inflows for these projects are Rs. 6,50,000, Rs. 95,000 and Rs. 1,00,30,000 respectively. Calculate the respective Profitability Index (PI) for the three projects.

SOLUTION:

The respective Profitability Index for the three projects would be as follows:

$$1. \quad PI = \frac{\text{Rs. } 6,50,000}{\text{Rs. } 5,50,000} = \underline{1.18}$$

$$2. \quad PI = \frac{\text{Rs. } 95,000}{\text{Rs. } 75,000} = \underline{1.27}$$

$$3. \quad PI = \frac{\text{Rs. } 1,00,30,000}{\text{Rs. } 1,00,20,000} = \underline{1.001}$$

It can be seen that in absolute terms, project 3 gives the highest cash inflows yet its Profitability Index is low. This is because the outflow is also very high. The Profitability Index factor helps us in ranking various projects.

J.W.

9.3 ESTIMATION OF PROJECT CASH FLOWS

Cash requirement of any project can be safely analyzed in terms of long term, short term, owned, borrowed, fixed cost, working capital, inflows and outflows. All this can be appropriately included in a projected cash flow statement. The process requires the following;

- **Identify the elements of cash flow:** A business typically will have three elements of cash flows: initial, operating and terminal. Initial Investment will be the capital expenditure and the contribution for net working capital to start the project. The operating one will comprise of the outflow and resulting inflow of the operations of the business. The terminal is the one what remains that is net inflow after paying off all the realizations of the assets on liquidation of the business when the economic life comes to an end. All the flows are post tax for the reasons that that element does not belong to the business or the owners.
- **How to estimate and basis of each such element:** There are four principles to be kept in mind. The first is the **separation principle** which means we have to separate the investment side from finance side which simply means separating assets for servicing cost of it. Cost flows in to investment side and the interest, if any, flows in to financing side. Then there is **incremental principle** which means estimate separately the cost which will be incurred even if the project is not run from the cost which we incur while running it. We can also say it is fixed versus variable cost. While considering the cash flows, considering the post- tax cash flows always, is advisable since the tax payments cannot be ignored. This is the **third principle**. If you ignore, you will have to discount pretax flows with a discount rate which may or may not be reliable. The last principle is **consistency principle**.
- **Collate all the components of cash flow:** All the components of the cash flow namely fixed capital, working capital, own capital (Equity), long term borrowing, outflow for operations such as purchases, production expenses, operating and administration expenses, selling expenses, interest cost, and inflow on account of sales and services are collated and net flow or the cash balance is found out to complete the cash flow statements. This is prepared for the period of the entire project period or for reasonably long enough period to reach a break even or achieve other objectives such as debt free status of closure or hive off etc.

9.3.4 Cost Analysis for Projects

What is the project cost? It is the cost incurred for setting up a project. Total project cost consists of total investment in the fixed assets that will enable the entity to start the production. If we divide it in to various sections, we have a break up as under:

Fixed assets: Fixed assets consist of all the tangible assets such as land, buildings, plant and machinery, electrical installations, office equipment, computers, vehicles, staff quarters, etc. Let's now see what constitutes the cost of some of these fixed assets;

- Land cost will be different for freehold land or lease land, the former being costlier because it gives lifetime rights to enjoy it. The cost will include stamp duty, legal fees, leveling and developing.
- Like land, the cost of factory building will vary based on whether is bought, constructed or on rental basis. Whatever it is, the cost of acquisition, stamp duty, legal cost, cost of construction and alterations or additions to the lease premises are some of the major elements of such cost.
- Plant and machinery will cost you as much as its purchase price, duties, freight inward, clearing and customs and installations expenses.
- Power line cost, internal cabling for connecting and electrifying the plant, transformers, sub-station, equipment and installations will be part of the proper power supply set up.
- Depending upon the nature of the plant activity, the company may be required to invest in environment clearance, maintenance and protections. Such cost is a fixed one and can include equipment for prevention of air and water pollution, planting of trees and creating clean and hygienic environment.
- The cost of other tangible fixed assets does not require elaboration.
- Apart from the above, the company may also have to invest in creating or acquiring initial intangible assets such as registrations, patents, trademarks, inventions, discoveries, data, software, systems etc.

Installation cost: Such cost will vary from industry to industry. A simple engineering unit will not incur installation cost as much as a refinery or pharmaceutical or a chemical process plant will be required. A heavy engineering unit or auto component unit may require installation of overhead cranes or goods movement facility. Moreover, apart from the cost, the installation requires deep planning including taking care of all the logistics and time and motion study. All direct cost including the cost of the manpower directly and wholly concerned with the installation will be accounted under this head.

Direct cost relating to setting up of the business: It is advisable to maintain records of expenses incurred by the company during construction period in such a way that the expense can be linked to the asset or a class of asset. At the end of the year or completion of the construction whichever is earlier, the expenses should be classified and be debited to the asset or class of the asset and credited to the Expenditure during Construction Period Account.

Other cost: This cost relates to general operating and administration cost the company incurs during the setting up of the project. This can also be capitalized appropriately among the capital assets that have been created. The basis has to be on sound accounting principles.

The analysis of this total cost: Although it is necessary that during the construction period there has to be continuous monitoring from various angles, the final cost analysis will be centering on the following points or issues.

The necessity: The analysis should reveal that the asset acquired was absolutely necessary; be it type, brand, size, capacity or power, and the numbers. This information can be found in the project report and final analysis should confirm the conformation.

The best combination: Many processes or methods of production have alternate suggestions and we need to find that the combination used is the best one.

Competitive bargaining: This means that the procurement process is so planned that the price paid for the asset is the best bargained one.

Cost versus recoverable duties: The duty structure of the country is such that there is not tax on the tax. In other words, the duty paid on an asset is allowed to be off set against the duties or taxes collected. This advantage is also called Input Tax Credit (ITC). Therefore, in the final analysis, it should be concluded that all the taxes paid on acquisition of the assets, to the extent recoverable, are not included in the cost of the asset.

Leveraging: There are many alternatives for fund raising whether it is for long term or for working capital. Similarly, with respect to assets too, one can go for buying or leasing. While carrying out the cost analysis, one should also consider that the company has taken into consideration optimal leverages while sourcing the finance and the assets. The ultimate object of such leveraging is to optimize the cost of operation and make the business more competitive or profitable.

Life span of the productive assets: While choosing the asset for business set up, the company will also take into account the life span of the asset to be acquired. If the life span is too long, the cost of repairs and maintenance will be high and the company will be saddled with an asset with outdated technology. If the life span is too short, the company will have to go for frequent replacement disrupting production. Based on the business plan and life cycle of the business, it will decide about the lifespan and the asset. This will also decide the depreciation rate and the period.

UNIT 10

CAPITAL BUDGETING FOR INTERNATIONAL PROJECT INVESTMENT DECISIONS

STRUCTURE

- 10.0 Objectives
- 10.1 Introduction
- 10.2 Foreign Investment Analysis
- 10.3 International Capital Budgeting
- 10.4 Issues Involved in Evaluation of Overseas Projects
- 10.5 Approaches for Evaluation of Overseas Projects
- 10.6 Evaluation Methods
- 10.7 Impact of Transfer Pricing

Let Us Sum Up

Keywords



International Portfolio Investment (IPI) and its constraints, focusing on the increasing trend of global investment:

- ✓ **Definition and Context:** IPI involves investing in foreign securities like treasury bills, mutual funds, and equities, often through cross-country remittance channels. In India, individuals can send up to USD 250,000 annually under the Liberalised Remittance Scheme (LRS), though this applies mainly to international mutual funds and listed stocks.
- ✓ **Benefits:** IPI offers access to diverse markets and products, providing valuable market information, stability in exchange rates, and reduced risk through diversification across countries.
- ✓ **Diversification:** Spreading investments across global markets lowers risk and enhances returns compared to domestic-only investments.
- ✓ **Barriers:** Challenges include capital flow restrictions, political or economic uncertainties (e.g., price controls, fiscal policies), and language barriers affecting information access.
- ✓ **Asset Allocation:** Effective IPI requires careful risk-return balancing, skillful security selection, and ensuring liquidity to meet redemption needs. Modern market mechanisms help identify high-rated securities.

International portfolio investment:

Direct and Indirect Channels: Domestic investments include equity, mutual funds (MF), debt, NPS, PPF, bank FDs, and SGBs. Internationally, options include ADRs, GDRs, foreign direct investing, global mutual funds, and ETFs. Indirect channels involve multinational corporations (MNCs), while direct channels use ADRs/GDRs or FDI for corporate funding. Online platforms also facilitate investment.

Exchange and Country Risk: Exchange risk arises from currency fluctuations due to local currency depreciation or appreciation, influenced by economic, political, or international events. Country risk involves political instability, legal systems, trade policies, and social turbulence, impacting investment stability.

Return and Risk of Foreign Investment: Returns depend on asset liquidity, risk, and economic stability. Liquid assets yield lower returns but are safer, while stable foreign exchange rates and reinvestment plans offer higher returns. International investments can outperform domestic ones but carry risks like political and economic instability.

Capital Asset Pricing Model (CAPM): CAPM links risk and expected returns, using beta to measure a security's volatility against the market. It helps price risky assets, though accurate future cash flow predictions would reduce the need for such models. Investors use CAPM for pricing and risk assessment.



Capital Asset Pricing Model (CAPM) for evaluating investment returns and risks:

CAPM Formula: Cost of Equity (r_e) = Risk-Free Rate (r_f) + Levered Beta (β_L) $\times$ Market Risk Premium ($r_m - r_f$).

The risk-free rate is the return on a no-risk security, and market risk premium is the difference between market return and risk-free rate. Beta measures a stock's volatility relative to the market (often S&P 500).

Arbitrage Pricing Theory (APT) for evaluating investment returns and risks:

APT Overview: APT is an alternative to CAPM, focusing on multiple macroeconomic factors (e.g., GDP growth, inflation, gold prices, stock indices) to explain asset returns and systemic risk. It helps identify mispriced securities for arbitrage opportunities.

International capital budgeting and project evaluation:

Capital budgeting for international projects involves assessing investments abroad, considering local markets for resources or consumption. It requires careful evaluation due to differing market dynamics.

Project Evaluation: This process determines a project's viability and impact by analyzing workflows, stakeholder value, and future patterns. For example, a US manufacturing plant with a \$10 million USD investment is evaluated, comparing the US risk-free rate (4%) and India's (7%).

Issues in Evaluation of Overseas Projects: While evaluation methods for overseas projects mirror domestic ones, a key challenge is:

Calculating Discount Rate: The risk-adjusted discount rate must be determined using local risk-free rates. For a project, the US risk-free rate is 4%, India's is 7%, and the company's required rate in India is 12%. The formula for the risk-adjusted rate is $(1 + r_a) = (1 + r_f) * (1 + r_p)$, where r_a is the risk-adjusted rate, r_f is the risk-free rate, and r_p is the risk premium.

UNIT 11

ADJUSTMENT OF RISK AND UNCERTAINTY IN CAPITAL BUDGETING DECISION

STRUCTURE

- 11.0 Objectives
- 11.1 Introduction
- 11.2 Sources & Perspectives on Risk
- 11.3 Sensitivity Analysis
- 11.4 Scenario Analysis
- 11.5 Hillier Model
- 11.6 Simulation Analysis
- 11.7 Decision Tree Analysis
- 11.8 Corporate Risk Analysis
- 11.9 Managing Risk
- 11.10 Project Selection under Risk
- 11.11 Risk Analysis in Practice

Let Us Sum Up



Overview of Capital Budgeting

Capital budgeting involves estimating the **cost** and **revenue** for a project, particularly for long-term investments. It includes factors like:

- Land, labor, materials, plant, and working capital costs.
- Revenue forecasting, which depends on market behavior, competition, and other external factors.

In international projects, risks multiply due to factors like:

- Exchange rate fluctuations.
- Foreign rules and regulations.
- Political instability and cultural differences.

These factors make the risk analysis crucial, as risks impact project costs, revenues, and profitability.

Types of Risks in Capital Budgeting

Capital budgeting risks can be:

Project-specific risks: Risks tied to the nature of the project, such as hazardous chemicals or political instability.

Market risks: Depend on demand, competition, and external policies.

Corporate risks: Include factors like management disputes or takeover bids (e.g., Future Retail, Indigo Airlines).

Examples of Risks:

A plant using hazardous chemicals may face shutdown due to environmental violations.

Companies expanding internationally face exchange rate risks and cultural misalignment.

Sensitivity Analysis

Sensitivity analysis evaluates how **changes in key variables** impact the project's NPV (Net Present Value) or IRR (Internal Rate of Return). This method identifies the **vulnerable aspects** of a project, helping decision-makers manage risks.

Steps of Sensitivity Analysis:

- ✓ **Identify Key Variables:** Such as investment cost, sales, variable costs, and fixed costs.
- ✓ **Mathematical Understanding:** Analyze how changes in variables impact NPV or IRR.
- ✓ **Perform Analysis:** Change one variable at a time to observe its effect.



11.6 SIMULATION ANALYSIS

Simulation means not real, imitation or deception. It is a process of creating a similar but artificial situation. What we do here is that firstly, we conduct analysis by finding out sensitivity of certain criteria of merit such as NPV, IRR, or any such criterion, to variation in basic factors. Then as a next step, we do simulations to find out likelihood under various scenarios to enable us to take the best suited decision. The steps involved can be briefly stated as under:

- Prepare model project report on expected lines.
- Calculate NPV. (it can be IRR too)
- Find out how the NPV is influenced by or related to the parameters and the exogenous variables.
- Parameters are input variables specified by the decision maker who is in charge of the project.
- Exogenous variables are those whose value is determined outside the model and imposed on the model. These are not in control of the decision maker. This may probably depend on random event. For example, if we use seeds, fertilisers and water and then depend on rains to give us say 100 tons of a crop; the seeds, fertilisers, water and yield are endogenous and the rainfall is the exogenous factor which is not influenced by other variables.
- Fix the values of the parameters and also estimate the probability of distributions of exogenous variables.
- Select a value at random from the probability distributions of each of the exogenous variables.
- Determine the NPV with relation to the randomly generated values of exogenous variables and the predetermined or specified parameter values.
- Simulate the above exercise to any number of times to generate large number of simulated NPV.
- Plot the frequency distribution of the NPV and then consider which simulation is most probable or suitable.

2. Sensitivity Analysis considering 2.5 % Adverse Variance in each variable

The table that follows demonstrates how to determine the impact on the NPV of the project in terms of percentages by adjusting one variable at a time by 2.5 percent (in a negative direction) while maintaining the other variables in their original states. As a result, it is clear that the change in selling price has the greatest impact on the NPV, accounting for 24.82 percent of the total change.

	Particulars	Base	Initial capital cost increased to Rs. 410 crore	Selling Price per Unit Reduced to Rs. 97.5	Variable Cost Per Unit increased to Rs. 51.25	Fixed Cost Per year increased to Rs. 51.25 crore	Units sold per year reduced to 4.875 crore
		(Rs.)	(Rs.)	(Rs.)	(Rs.)	(Rs.)	(Rs.)
A	Selling price perunit	100	100	97.5	100	100	100
B	Variable cost perunit	50	50	50	51.25	50	50
C	Contribution perunit (A - B)	50	50	47.5	48.75	50	50

(Cont.)

	Particulars	Base	Initial capital cost increased to Rs. 410 crore	Selling Price per Unit Reduced to Rs. 97.5	Variable Cost Per Unit increased to Rs. 51.25	Fixed Cost Per year increased to Rs. 51.25 crore	Units sold per year reduced to 4.875 crore
		(Rs. in Cr.)	(Rs. in Cr.)	(Rs. in Cr.)	(Rs. in Cr.)	(Rs. in Cr.)	(Rs. in Cr.)
D	Number of units sold per year (units in Crores)	5	5	5	5	5	4.875
E	Total Contribution (C × D)	250	250	237.5	243.75	250	243.75
F	Fixed cost per year	50	50	50	50	51.25	50
G	Net Cash Inflow per year (E - F)	200	200	187.5	193.75	198.75	193.75
H	PV of Net cash Inflow per year (G × 2.673)	534.60	534.60	501.19	517.89	531.26	517.89
I	Initial capital cost	400	410	400	400	400	400
J	NPV (H - I)	134.60	124.60	101.19	117.89	131.26	117.89
K	Percentage Change in NPV	-	-7.43%	-24.82%	-12.41%	-2.48%	-12.41%

Scenario Analysis: Explanation and Application

Scenario analysis is an extension of **sensitivity analysis**, where instead of changing one variable at a time, multiple variables are changed simultaneously to explore different outcomes. This method evaluates the financial implications of three potential scenarios:

- **Pessimistic Scenario:** Assumes worst-case values for all variables.
- **Expected Scenario:** Uses normal or average values for variables.
- **Optimistic Scenario:** Assumes best-case values for all variables.

In scenario analysis, all variables are assumed to move in the same direction at the same time (e.g., if sales increase, variable costs and profits will also increase). This approach gives a clearer picture of potential outcomes when many factors change together, offering a more holistic view of risks and rewards.

Example: XYZ Ltd. Project Analysis

The table presented in the question outlines a scenario analysis for XYZ Ltd.'s project, showing possible outcomes based on different cash flow assumptions over a 3-year period. The company considers three scenarios to evaluate the project's net present value (NPV):

Particulars	Worst Case	Most Likely	Best Case
Year 1 Cash Flow	₹450,000	₹550,000	₹650,000
Year 2 Cash Flow	₹400,000	₹450,000	₹500,000
Year 3 Cash Flow	₹700,000	₹800,000	₹900,000

Here are the NPVs for each scenario:

1. **Worst Case NPV:**

Cash Flows: ₹450,000 (Year 1), ₹400,000 (Year 2), ₹700,000 (Year 3)

NPV: ₹-110.15

2. **Most Likely NPV:**

Cash Flows: ₹550,000 (Year 1), ₹450,000 (Year 2), ₹800,000 (Year 3)

NPV: ₹100.85

3. **Best Case NPV:**

Cash Flows: ₹650,000 (Year 1), ₹500,000 (Year 2), ₹900,000 (Year 3)

NPV: ₹311.85

Analysis with Worst Case Scenario for Year 3

XYZ Ltd. is confident about the cash flows for the first two years but uncertain about Year 3. To mitigate risk, they assume the **worst-case cash flow** for Year 3 while keeping the other two years as expected. The NPV is calculated as:

$$NPV = \text{Initial Investment} + \left(\frac{\text{Cash Flow Year 1}}{(1 + 0.09)^1} \right) + \left(\frac{\text{Cash Flow Year 2}}{(1 + 0.09)^2} \right) + \left(\frac{\text{Cash Flow Year 3 (Worst Case)}}{(1 + 0.09)^3} \right)$$

For this calculation, the NPV comes out to ₹23,871.

Conclusion and Interpretation

Scenario analysis provides a more comprehensive understanding of a project's financial viability by evaluating various possible outcomes:

- **Worst Case Scenario:** The NPV is negative (₹-110.15), signaling that the project might not be profitable if cash flows fall short.
- **Most Likely Scenario:** The NPV is positive (₹100.85), indicating that the project is expected to break even or generate a moderate return.
- **Best Case Scenario:** The NPV is the highest (₹311.85), suggesting a strong return on investment if things go well.

11.3 SENSITIVITY ANALYSIS

The uncertainty of future movements of certain variables can move either way, that puts all our present estimates and the project or investments to jeopardy. We need to safeguard against this. The variables and important components of capital budget are cost, revenue and net profits. Taking all possible variables into account will be a difficult task, making calculations complex. Sensitivity analysis aims to assess the impact of changes in each of these important variables on our projections or estimates. To explain the sensitivity analysis due to such variations, let's take a simple example of a project to be implemented. Table A below gives brief but important details of a PET Tube manufacturing project.

Table A

Rs. in Lakhs

	Year 0	Years 1-10
Investment	(40,000)	
Sales		36,000
Variable Costs		24,000
Fixed Costs		2,000
Depreciation @10% of project cost		4,000
Profit before Tax		6,000
Tax @ 33.33%		2,000
Profit after Tax		4,000
Cash flow from operations		8,000
Net cash flow	(40,000)	8,000

On a total investment of Rs. 40,000 lakhs, targeted sales are Rs.36,000 lakhs. The PAT (Profit After Tax) is Rs.4,000 lakhs resulting into net cash flow from operations at Rs.8,000 lakhs. Considering discounting rate of 12% and project term as 10 years, the Net Present Value (NPV) works out to Rs.5,200 lakhs. This is a normal scenario without getting in to risk arising out of uncertainty. Now if we consider the best and the worst scenarios in the case of investment, sales, variable cost and fixed cost; the results give you surprises. To calculate the impact of changes in one variable, we presume that the other variables remain constant.

The Table B below shows that in the worst scenario, if the investment cost goes up to Rs.48,000 lakhs, the NPV nose dives to -1300 (negative). Similarly, if the sales fall to Rs.30,000 Lakhs, the NPV falls to -2340(negative).

Table B

	Rs. (Lakhs)			Rs. (Lakhs)		
	Range			NPV		
Key Variable	Pessimistic	Expected	Optimistic	Pessimistic	Expected	Optimistic
Investment	48,000	40,000	36,000	-1,300	5,200	8,440
Sales	30,000	36,000	42,000	-2,340	5,200	12800
Variable cost as % of sales	70	66.67	65	680	5,200	7460
Fixed Costs	2600	2000	1600	2940	5,200	6706

We can, therefore, say that the sensitivity analysis is a good method of assessing the risk, enabling us to find out how robust or weak the project is. If we find such impact and identify the vulnerable aspects, we can change our plan or make amendments to contain the risk. This method can thus be employed to evaluate the budget and the project to a higher degree of risk management. The limitation of this method is that it only shows the NPV and only one variable is changed at a time. To an extent this method is also subjective because it depends on the perception of the decision maker of the analyst.

Steps involved in Sensitivity Analysis

The steps that need to be taken in order to perform a sensitivity analysis are as follows:

1. Identifying the factors that have an impact on the NPV (or IRR) of the project.
2. Developing a mathematical understanding of the connections between the different variables.
3. Conducting an analysis to determine how the changes in each of the variables will affect the net present value (or internal rate of return) of the project.

ILLUSTRATION 1

A Ltd. is considering its new project with the following details:

Sr. No.	Particulars	Figures
1	Initial capital cost	Rs. 400 Cr.
2	Annual unit sales	5 Cr.
3	Selling price per unit	Rs. 100
4	Variable cost per unit	Rs. 50
5	Fixed costs per year	Rs. 50 Cr.
6	Discount Rate	6%

Required:

1. Calculate the Net Present Value (NPV) of the project. Assume that the tax rate is zero.
2. Compute the impact on the project's NPV considering a 2.5 per cent adverse variance in each variable. Which variable shall have the maximum effect?

Consider Life of the project as 3 years.

11.10 PROJECT SELECTION UNDER RISK

A project report, once ready, will tell us about what we can expect from the implementation which may include an NPV or IRR or any other criterion of merit. Further analysis will also reveal the major deviations under various scenarios, finally leading us to either go ahead with or reject the project. Thus, we face a situation where we have to take a decision under various risk probabilities. Under such circumstances, we follow the following process:

- a) **Evaluation using our judgemental capabilities;** though highly subjective, this method is generally the first to be resorted to. We are always proud and confident of our senior management which is generally a pool of talented executives having worked for decades in responsible positions, steering companies to success. It may be a project evaluation committee or budgeting committee or senior executive committee or the Board itself. No special methods or complicated calculations are deployed. It's just brain storming. Some may be highly optimistic, some pessimistic, some having pecuniary interest or someone a devil's advocate. All will debate and will be heard. Under discussion will be pros and cons, risk and rewards, side effects, collateral damages, competitor's response, capital constraints, reputation risk and host of things. It will be free for all but finally a judgemental all things considered. This is what we in our personal life do choosing a career or a spouse!
- b) **How quickly we become risk free:** This is an important aspect of any large or risky project. We are always eager to be risk free which indirectly means when we get our money back. Larger the risk, shorter the period we would like to get out of it because longer the term, uncertainties increase and new unexpected events upset our plans. In other words, we presume that risk is a function of time. Therefore, the payback period determination guides us about acceptability of a project. By following this method for decision making, we are trying to control the risk by bringing it closer to reality and also bring an end to it. The opponents of this theory, particularly in investments in securities, argue that it is in the long term that a risk is evened out and the one who stays put has the last laugh. Hence this method may not be the only or the best method.
- c) **Risk adjusted discount rate:** When we make an investment in a project, we try to first find out what will be the return on it. For that purpose, we calculate NPV which has to be positive to fund the project. How do we work out NPV? We use a discount rate to measure the NPV. In the normal circumstances, if we do not anticipate any risk, we will adopt the average cost of capital as the discount rate. However, if we find that there is a possibility of certain risk occurrence, we need to provide for the same and we enhance the rate. This rate is the risk adjusted discount rate. We calculate the same in the following manner.

UNIT 12

DECISION MAKING

STRUCTURE

- 12.0 Objectives
- 12.1 Introduction
- 12.2 Decision Making using Cost-Volume-Profit (CVP) Analysis
- 12.3 Decision Making using Relevant Cost Concepts
- 12.4 Decision Making using Activity Based Costing (ABC)
- 12.5 Ethical and Non-Financial Considerations Relevant to Decision Making

Let Us Sum Up



Financial Decisions Beyond Capital Budgeting

Financial decisions in an organization are not limited to capital budgeting but extend to various strategic and operational areas, such as:

- ❖ Product/service selection.
- ❖ Pricing strategies.
- ❖ Make-or-buy decisions.
- ❖ Accepting or rejecting special orders.

These decisions require a mix of data analysis, estimates, and management preferences. **Ethics, social responsibility, and environmental considerations** often play a role, emphasizing that financial decisions are not solely profit-driven (e.g., installing a pollution control unit for environmental benefits).

Cost-Volume-Profit (CVP) Analysis as a Decision-Making Tool

CVP analysis focuses on the relationship between **cost**, **volume**, and **price** to analyze profitability and breakeven points. Key insights include:

- **Cost Dynamics:**

Fixed costs decrease per unit with higher production volumes.

Variable costs vary with production but depend on product mix and processes.

Economies of scale often reduce costs but may require incremental investments.

- **Impact of Volume and Pricing:**

Higher production volume can increase absolute profits but may entail higher marketing costs.

Pricing affects profitability and determines the breakeven point, influencing the financial health of the business.

Factors Affecting CVP and Investment Decisions

Key considerations:

- **New Product Launches:** High initial costs and low volumes delay breakeven.
- **Consumer Products:** Large markets require significant investment in production and branding, affecting cost structures and payback periods.
- **Specialty Products:** High margins but require heavy R&D and branding expenses.

Interdependence of Factors: Fixed/variable costs, pricing, and market conditions are interrelated, and any change in one factor necessitates adjustments in others.

External Influences on CVP Analysis

CVP analysis must account for unforeseen external events, such as:

- Economic disruptions (e.g., pandemics, wars).
- Regulatory changes (e.g., trade treaties, quotas).

Applications of CVP Analysis

CVP analysis aids decision-making by answering questions like:

- How will profitability change with reduced prices but increased volume?
- What additional sales volume is needed to cover increased fixed costs (e.g., from an advertising campaign)?
- What is the best compensation structure for sales staff (fixed salary vs. commission)?
- How will fluctuations in sales, prices, or costs impact financial outcomes?

Illustration 1

A company manufactures cycles for both adults and children. Given below is information about cycles made for children—

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Particulars	Traditional CVP Analysis	Activity Based CVP Analysis
Monthly Demand and Production	10,000 units	10,000 units
Selling Price	Rs. 8,000 per unit $\times 10k$	Rs. 8,000 per unit
Variable Cost per unit	Rs. 7,500 per unit $\times 10k$	Rs. 7,500 per unit
Fixed Cost p.m. (as identified under each cost system)	Rs. 10,00,000 p.m.	Rs. 8,00,000 p.m.

In the context of a traditional CVP study, fixed costs are defined as costs that do not change in proportion to the amount of product or service being purchased. After conducting an analysis using activity-based costing, it was determined that the monthly fixed costs that do not change, based on factors such as volume or any other cost driver, are Rs. 8,00,000 per month. According to the findings of the analysis, a milling machine is utilised to cut metal into steer support. The production of these steer supports occurs in batches of twenty-five units. After completing a batch intended for the children's cycle, the following

batch would be intended for the adult cycles. As a result, there would be a change in the set-up following the completion of each batch. If 10,000 children's bicycles need to be manufactured, the needed number of set-ups is equal to 10,000 steer support divided by 25 each batch. This results in a total of 400 set-ups. Each set-up costs Rs. 500, which includes charges for items such as changing the oil and purchasing jigs, among other things. In the past, the standard CVP analysis combined this cost with the fixed cost altogether.

You are required to

- (i) Determine the break-even point and profit per month using both the classic(traditional) CVP method and the Activity Based CVP method.
- (ii) As the manager of the plant, it is in your best interest to limit the amount of times that the machine needs to be set up so that it can operate at full capacity. Imagine for a moment that the milling machine has the capability of producing different types of cycles, such as adult cycles, sports cycles, and so on at any time. As a result, you suggest that the batch size of the children's steer support be increased to 50 individual units in a single batch. The current number of set-ups, 400 (10,000 units divided by 25 units), will be reduced to 200 (10,000 units divided by 50 units). The fabrication of larger batches will necessitate the leasing of more inventory storage space, which will result in an additional expense of Rs. 50,000 per month for the business. Analyse the impact on BEP (units per month) and PPM (earnings per month).
- (iii) When should the cost of labour be included in the overall estimate of how much a set-up will cost? Explain.
- (iv) How shall the number of set-ups on the milling machine and the cost of each set-up affect the machine's overall flexibility? Explain.

Solution:

- (i) (a) Break-even point (units per month) and profit per month under classical(traditional) CVP analysis:

Selling Price per unit	Rs. 8,000
Variable Cost per unit	Rs. 7,500
Contribution per unit	Rs. 500
Fixed Cost per month	Rs. 10,00,000
Break-even Point (per month in units) = Fixed Cost p.m. / Contribution p.u. = Rs. 10,00,000 / Rs. 500 per unit	2,000 units
Monthly Demand (units)	10,000 units
Profit per month = {Monthly demand (units) × Contribution per unit} – Fixed Cost per month = (10,000 × Rs. 500 per unit) – Rs. 10,00,000	Rs. 40,00,000

- (b) Break-even point (units per month) and PPM (profit per month) under Activity Based CVP method. Number of units produced per batch is 25. Therefore, number of set-ups will be 10,000 units / 25 units = 400 per month.

Selling Price per unit	Rs. 8,000
Variable Cost per unit	Rs. 7,500
Contribution per unit	Rs. 500
Fixed Cost per month (per Activity Based method)	Rs. 8,00,000
Break-even Point (per month in units) = {Fixed Cost p.m. + (number of set-ups × cost per set-up)} / Contribution p.u. = {Rs. 8,00,000 + (400 × Rs. 500 per set-up)} / Rs. 500 per unit = Rs. 10,00,000 / Rs. 500 per unit	2,000 units
Monthly Demand (units)	10,000 units
Profit per month = {Monthly demand (units) × Contribution per unit} – (Fixed Cost per month + Set-up cost per month) = (10,000 × Rs. 500 per unit) – (Rs. 8,00,000 + Rs. 200,000) = Rs. 50,00,000 – Rs. 10,00,000	Rs. 40,00,000

Despite the fact that the BEP units and the earnings each month are identical when using either method, the Activity-Based method has brought to light the fact that there are 400 set-ups being conducted each month. This would provide management with additional data to work with in order to enhance the functioning of the operations.

- (ii) Break-even point (units per month) and profit per month under Activity Based CVP analysis: Batch size increased from 25 to 50 units; monthly set-ups reduce from 400 to 200 per month.

Selling Price per unit	Rs. 8,000
Variable Cost per unit	Rs. 7,500
Contribution per unit	Rs. 500
Fixed Cost per month (per Activity Based method) Additional cost p.m. for inventory storage = Rs. 50,000	Rs. 8,50,000
Break-even Point (per month in units) = {Fixed Cost p.m. + (number of set-ups × cost per set-up)} / Contribution p.u. = {Rs. 8,50,000 + (200 × Rs. 500 per set-up)} / Rs. 500 per unit = Rs. 9,50,000 / Rs. 500 per unit	1,900 units
Monthly Demand (units)	10,000 units
Profit per month = {Monthly demand (units) × Contribution per unit} – (Fixed Cost per month + Set-up cost per month) = (10,000 × Rs. 500 per unit) – (Rs. 8,50,000 + Rs. 1,00,000) = Rs. 50,00,000 – Rs. 9,50,000	Rs. 40,50,000

Decision Making Using Activity-Based Costing (ABC)

Activity-Based Costing (ABC) is a method used for determining costs that aid in decision-making processes like pricing, outsourcing, and profitability analysis. It focuses on activities as the fundamental cost objects and assigns costs to products or services based on their consumption of activities.

Traditionally, costs were allocated using broad measures like production volume, which often led to inaccurate cost allocation. ABC improves accuracy by identifying and assigning costs based on the actual activities that drive them.

Key Concepts in ABC

1. Activity-Based Allocation

Identifies various activities (e.g., quality control, material handling) within the business.

Allocates costs to products/services based on their consumption of these activities.

2. Cost Drivers

Factors that cause costs to change for an activity.

Examples: Labor hours, machine hours, number of purchase orders.

○ Types:

- ❖ **Resource Cost Drivers:** Measure resource consumption by an activity.
- ❖ **Activity Cost Drivers:** Measure activities required for production or services.

3. Cost Pools

Grouping of costs associated with a specific activity (e.g., costs of quality checks).

4. Cost Objects

The item (product, service, customer) for which cost measurement is required.

Steps in Implementing ABC

- ✓ **Identify the Project:** Define the specific costing exercise.
- ✓ **Identify Activities:** Subdivide the production or service process into measurable activities.
- ✓ **Identify Direct Costs:** Allocate costs directly traceable to products or services.
- ✓ **Relate Overheads to Activities:** Attribute indirect costs to the respective activities.
- ✓ **Distribute Support Services:** Allocate support service costs (e.g., HR, maintenance) across primary activities.
- ✓ **Determine Cost Drivers:** Assign a measurable driver for each activity.
- ✓ **Calculate Total Cost:** Compute the total cost for each product or service.

Applications of ABC in Decision-Making

1. **Product Pricing:**

Helps set competitive and profitable prices by understanding the true cost of activities.

2. **Profitability Analysis:**

Determines profitability at the product, service, or customer level.

3. **Outsourcing Decisions:**

Assesses whether outsourcing specific activities or functions is cost-effective.

4. **Improving Efficiency:**

Identifies high-cost activities, enabling management to optimize processes.

Advantages of ABC

- ✓ Provides detailed insights into cost structures.
- ✓ Increases cost allocation accuracy.
- ✓ Highlights process inefficiencies.
- ✓ Supports strategic decision-making with reliable cost data.

Conclusion:

ABC is a robust tool for understanding and managing costs effectively. By focusing on activity-level cost drivers, it enhances managerial decisions, leading to better pricing, cost control, and profitability analysis.

Ethical Decision-Making in Finance

The decisions based on ethics are driven by a different set of considerations, such as:

- **Is it the truth?** The company should always stick to the Truth. When you speak a truth, you don't have to remember what you have said because the truth, as fact, will always remain stored in your memory and will come out whenever so required. On the other hand, a lie will always require a cover up. A small slip and you will be exposed. Thus, it is always better to write, speak and express truth and truth only
- **Is it a breach of trust?** If someone has trusted you, never misplace it. Trust is a foundation of many relations, including business relations and dealings. Breach of trust can be a legal ground for making a contract null and void. A criminal breach of trust can land you in jail. A party, who is not trustworthy, will lose all its goodwill and consequently, will be out of business.
- **Is it respectful?** An organisation should always ensure that it is respectful of all the men and matters. An employee, a colleague, a customer, or for that matter anyone, with whom you have to deal, should be respected. Everyone has his own status and a place in the eyes of the people around him. Do not brush him aside saying that he needs our business and earns fortune through us. Respect begets respect and hence it will be beneficial for our own business.
- **Is it responsible?** All of us have to act as a responsible person or citizen. The word responsible means having an obligation to do something, or having control over or care for someone as a part of one's own job. Hence, while taking decisions, we should act with sense of responsibility. If we do so, all around will benefit and we will add to our goodwill.
- **Is it fair?** We should not always look at the things from the view point of our advantage. Or that is it advantageous to us? To be a successful, as well as reputed company, we should adopt policy of fairness in our dealings. Our action should be such that fairness is visible through it. Fairness means treating people or matter with equality without any favouritism or discrimination. For example, when we constitute a committee for employee welfare, we should ensure that every segment of workforce is fairly represented on it. When we are negotiating wage revision, we should always keep in mind the views, feelings and requirements of the workers. Even while dealing with competitors, the element of fairness should not be given a go by, under any circumstances. In fact, fairness should be in our nature and blood and this policy will go a long way for the success of the business.
- **Is it compassionate?** Compassion is a fine human quality. It is a feeling or showing a sympathy and concern for the fellow beings. We should not hesitate taking certain decisions on compassionate ground when the rules and procedures do not cover certain situations. An example can be giving extended maternity leave to a female employee whose child is born with certain problems and the mother needs some more time to attend to it. Or, giving extra increment to an employee, who has a bright child and needs extra money for her education

MODULE – C

VALUATION, MERGERS & ACQUISITIONS

- Unit 13. Corporate Valuation
- Unit 14. Discounted Cash Flow Valuation
- Unit 15. Other Non-DCF Valuation Models
- Unit 16. Special Cases of Valuation
- Unit 17. Merger, Acquisition & Restructuring
- Unit 18. Deal Structuring and Financial Strategies





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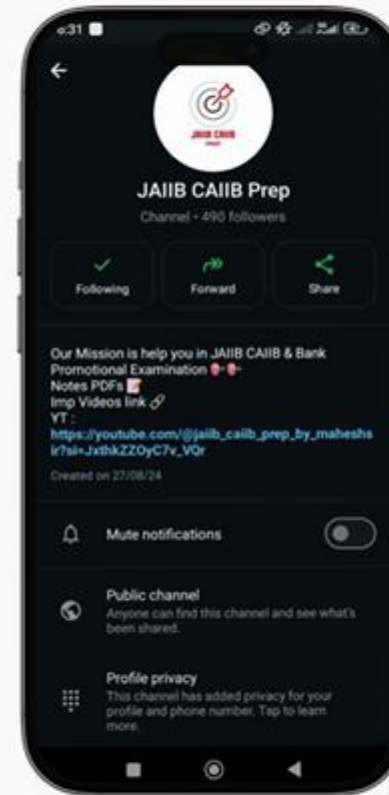


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